“Room Temperature Humidity Control & System Microcontroller-Based Room”

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Abstract :

Microcontroller-based humidity, Temperature , and Lighting control system for boarding house comport and efficiency. In a boarding house environtment, problems often arise related to comport and efficiency, such as the need for lighting on the terrace and humidity in the bathroom. These problems can be solved with a microcontroller-based automatic system that integrates LDR (Light Dependent), PIR (Passive Infrared), and DHT (Digital Humidity and Temperature) sensors.

Keywords : Microcontroller, ESP-32, LDR Sensor, PIR Sensor, DHT Sensor, Humidity, Temperature, Lighting, Efficiency, Comfort, Boarding House.

I. INTRODUCTION

As a boarding student, we are often with a variety of problems that interfere with the comfort and efficiency of daily life. Some of the problems that often experienced include the need for lighting on the terrace of the boarding house when coming home late at night and the problem of moisture in the bathroom that causes moldy walls. Night and moisture problems in the bathroom that cause moldy walls and peeling paint, both of these problems not only cause inconvenience. But also cause unnecessary waste of electricity. The need for optimal lighting in the boarding house terrace is crucial for the safety and comfort of the occupants, especially when returning home at night. Without adequate lighting , boarders have to rely on light sources that inefficient or unsafe. Even unsafe, lighting that stays on all night is also not ideal solution as it increases electricity consumption and costs. Solution as it will increase electricity consumption and operational costs , on the other hand moisture problem in the bathroom is an issue that is often faced by boarders. High humidity can cause mold growth on the walls and damage to the paint finish and damage to the paint layer. This not only ruins the aesthetics of the bathroom but can also cause health problems for residents, such as allergies or respiratory problems. This condition also result in the need for more frequent maintenance and repairs more frequently, which ultimately increases the cost of the bathroom.

These problems often arise because of the busyness of the boarders who sometimes forget or are lazy to organize lighting and ventilation. Boarders who sometimes forget or lazy to adjust lighting and ventilation manually. Pilling up college assignments, extracurricular activities , and the other daily activities make boarders pay less attention to the management of their living environment. Their living environment, to overcome these problems, we plan to create an automatic microcontroller-based system that can control humidity, temperature, and lighting using a can control humidity by using the LDR (Light Dependent Resistor), Sensor Pir (Passive Infrared) and Sensor DHT (Digital Humidity and Temperature). The selection of these sensors is based on the functionality and efficiency offered. With this automatic system, it is expected to reduce electricity wastage and increase the comfort of the boarders. Electricity and increase the comfort of boarders, this system also provides practical solution for busy boarders, ensuring the boarding environment remains comfortable and healthy without requiring extra attention and effort from the boarders. Comfortable

and healthy without requiring extra attention and effort from the residents.

II. PROJECT CONCEPT

Microcontroller-based Room Temperature Humidity Control System is a technology project designed to control and monitor environmental conditions in a room automatically using a microcontroller. As a boarders, we are often faced with various problems that interface with the comfort and efficiency of daily life. Some of the problems that are often experienced include the need for lighting on the terrace of the boarding house when returning home late at night and moisture problems in the bathroom that cause moldy walls and peeling paint. To solve these problems, we plan to create microcontroller-based automatic system that can control humidity, temperature, and lighting using LDR (Light Dependent Resistor), Sensor Pir (Passive Infrared) and Sensor DHT (Digital Humidity and Temperature). The selection of these sensors is base on their functionality and efficiency.

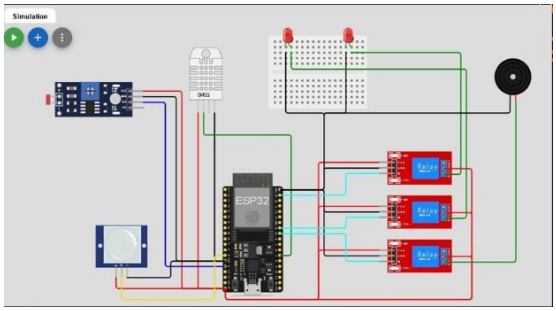
a. Components Used
    - ESP-32
    - DHT 22 Sensor
    - PIR Sensor
    - LDR Sensor
    - Relay Module
    - LED
    - Jumper Cable
    - Fan
    - Step Up

b. Steps

1. PIR Sensor
    - VCC is connected to 5V ESP32.
    - GND is connected to ESP32 GND.
    - OUT is connected to GPIO 15 of ESP32.
2. LDR Sensor
    - One leg of the LDR is connected to 3.3V of ESP32.
    - The other leg is connected to a 10k Ohm resistor, and the center point of the resistor to the GND ESP32.
    - The point between the LDR and the resistor is connected to GPIO 34 of ESP32.
3. DHT 22 Sensor
    - VCC is connected to 3.3V ESP32.
    - GND is connected to ESP32 GND.
    - DATA is connected to GPIO 4 of ESP32.
4. Relay Module .

- VCC is connected to 5V of ESP32.
- GND is connected to GND ESP32.
- IN1 is connected to GPIO 25 of ESP32 (bathroom light).
- IN2 is connected to GPIO 26 ESP32 (porch light).
- IN3 is connected to GPIO 27 ESP32 (fan).

c. Project Simulation



In the simulation above, the fan component part is replaced by a buzzer due to simulation limitations.

III. RESULT & DISCUSSION

Room Temperature Humidity Control System and Room Control Based on Microcontroller is a technology project designed to control and monitor environmental conditions in a room automatically using a microcontroller. Using a microcontroller, The above program is designed to control humidity, temperature, and lighting in a boarding house using a microcontroller based microcontroller integrated with several sensors and actuators. This system utilizes DHT22 sensors to measure air temperature and humidity, PIR sensors to detect human movement, and LDR sensors to detect human movement. sensor to detect human movement, and LDR sensor to detect light levels. These three sensors work together to control three output devices: bathroom lights, patio lights, and fans. bathroom light, porch light, and fan, through a relay module. This program uses several main components connected to the ESP32 pins as follows.

as follows:

a. PIR Sensor (Pin 12):

Used to detect human movement around the area that is being monitored.

b. LDR Sensor (Pin 13):

Uses this pin to detect the lighting level in the surrounding environment, providing a digital output.

c. DHT22 Sensor (Pin 4):

Used to measure the temperature and humidity of the air in a room.

d. Relay Module: Controls electrical devices, with each relay

connected to:

- Bathroom Light (Pin 23)
- Patio Light (Pin 19)
- Fan (Pin 18)

Then in the setup() function, the pins for sensors and relays are configured as inputs or outputs as needed. as inputs or outputs as needed. The DHT22 sensor is initialized to start reading temperature and humidity data. Serial communication is started with a baud rate of 115200 for monitoring and debugging the sensor data displayed on the Serial Monitor. All relays are initialized in the off state to ensure the connected devices do not turn on when the system is newly powered on. connected devices do not turn on when the system is just started up.

Then in the loop() function operation, data from each sensor is read and processed every two seconds. First, the value from the PIR sensor is read to detect whether there is human movement around the sensor area. If there is movement (low PIR value), the bathroom light is turned on; if there is no motion, the bathroom light is turned off. Second, the value from the LDR sensor is read to determine the lighting level in

the environment. If it is dark (low LDR value), the porch light is turned on; if it is In bright conditions, the porch light is turned off. Third, the temperature and humidity values from the DHT22 sensor are read and displayed on the Serial Monitor for monitoring. If the humidity exceeds 70%, the fan is turned on to reduce humidity; if the humidity is below or equal to 70%, the fan is turned off.

IV. CONCLUSION

The program for the project "Room Temperature Humidity Control System and Microcontroller-Based Room Control" above successfully implemented an automatic system to control lighting and humidity in a boarding house using an ESP32 microcontroller integrated with LDR, PIR, and DHT22 sensors. LDR, PIR, and DHT22 sensors. This system is designed to improve efficiency, comfort, and health of the boarding house environment by automatically adjusting the lighting and ventilation based on the environmental conditions and ventilation automatically based on the environmental conditions detected by the sensors.